

Who Profits When a New Kind of Offering Appears?

Evidence from 3.1 Million US Trademark Registrations

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Abstract

A new kind of offering creates value; whether its creator keeps that value depends on how easily it is imitated and on who holds the assets needed to sell it (Teece, 1986). We measure both sides on one outcome for every entrant: whether a registered US trademark’s goods are still in commercial use at the sworn five-year declaration. Each of 3.1 million registrations from 2002–2018 is scored on two positions of its goods-and-services language within its industry: atypicality, how unusual the language is, and lead, whether the industry later moved toward it. Unusual offerings create value that lasts: the most unusual filing in a class and year is still in use 5.4 points more often than the most typical, and renewed at year ten 4.3 points more often. Early offerings lose it: the most leading is in use 2.2 points less often than the most lagging. Who keeps the value follows the appropriability account. Patents double the advantage of unusual language and do nothing for the cost of being early. Growth in entry and the geographic concentration of rivals halve the advantage. The cost of leading falls on filers without counsel and on platform businesses, and it is largest in a dozen industries, from tobacco to technology services, where it lasts to renewal.

Keywords: profiting from innovation; first-mover advantage; appropriability; complementary assets; trademarks; product survival.

1 Introduction

A new kind of offering creates value, and the firm that brings it to market often does not keep it. Where an offering is easy to imitate and the assets needed to sell it belong to others, imitators and the holders of those assets capture what the innovator created; where it can be protected, or the innovator holds the assets itself, the innovator keeps it (Teece, 1986). The question of who profits is the one beneath a longer argument about entry order.

Pioneers keep the markets they open. In surveys of consumer-goods businesses, market pioneers held larger shares than early followers, and early followers larger than late entrants (Robinson & Fornell, 1985). A pioneer can stay ahead in technology and learning, take scarce inputs before others arrive, and hold customers who would pay to switch (Lieberman & Montgomery, 1988); when a new industry is later winnowed, the early entrants are the likeliest to survive the shakeout (Klepper, 1997; Klepper & Miller, 1995).

Most pioneers fail, and followers take what they proved. The histories of fifty product categories showed that about half of the true first entrants failed and that the firms remembered as pioneers had usually entered later (Golder & Tellis, 1993); in consulting, rivals copy new offerings routinely (Semadeni & Anderson, 2010). Whether moving first pays depends on the resources the entrant

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brings (Lieberman & Montgomery, 1998) and on how fast technology and demand change (Suárez & Lanzolla, 2007).

Creating a new category pays. A practitioner literature holds that firms which create uncontested market space, often by combining attributes from different industries, grow faster and earn more than those that compete in existing categories (Kim & Mauborgne, 2004).

These claims are argued from cases, most of them firms that survived to be studied. The trademark record covers every entrant, including those that failed, and dates what each one offered. A US trademark registration carries a description of the goods or services the mark covers. Between the fifth and sixth year after registration the owner must swear that the mark is still in commercial use on those goods, or the registration is cancelled; the declaration therefore records whether the product was still being sold. We score each registration’s description on two positions within its industry (Silver, 2026): *atypicality*, how unusual the language is against the filings before and after it, and *lead*, whether the industry’s later filings moved toward it. Atypicality measures a new kind of offering in its industry: under a theme model fitted across all industries, it largely records language that brings to an industry what others describe. Lead measures arriving where the industry was about to go.

Four results follow, organized by the appropriability account. First, unusual offerings create value that lasts, and early ones lose it: the most unusual filing in its class and year is still in use 5.4 points more often than the most typical and the most leading 2.2 points less often than the most lagging, and both effects hold at the ten-year renewal. Second, imitation erodes the value: the advantage of unusual language halves where a class’s filing volume grows fast and where the filers of the same theme concentrate in place. Third, protection keeps it: owners who patent keep 7.6 points of the advantage against 3.7 for owners who do not, and patents do nothing for the cost of being early. Fourth, the cost of being early falls where the entrant lacks complementary assets: on filers without counsel, on platform businesses, and in a dozen industries where it lasts to renewal.

Section 2 describes the registrations, the two positions and the factors. Section 3 reports the value created and lost. Section 4 compares every factor on one scale. Section 5 reports what erodes the value of unusual offerings, and Section 6 what lets the creator keep it. Section 7 asks who collects within a wave of entry, and Section 8 whether industry structure predicts where early entrants lose. Section 9 builds the geographic measure used throughout, Section 10 maps the results by industry, and Section 11 returns to the question of who profits.

2 Registrations, positions and factors

2.1 Registrations and outcomes

The unit is the registration: 3,104,534 US trademark registrations issued 2002–2018, each counted once in its lowest Nice class (the international classification’s 45 classes of goods and services). The main outcome is survival of the five-year proof: a registration passes unless it was cancelled for want of the declaration between 4.0 and 8.5 years after registration. 47.7% of registrations pass. The second outcome is renewal at year ten among registrations that passed the proof, observed for the 871,819 registrations of 2002–2013 that reached a terminal status; 59.9% were renewed. For the capital-markets outcomes, the unit is the owner’s first registered filing, linked by name to private-offering notices and to listings (Silver, 2026).

2.2 Two positions of the language

A topic model reduces every goods-and-services description to a mix of fifty themes. For each filing, the mix is compared by Kullback–Leibler divergence with the mix of its own class’s filings in the five years before its filing date and in the five years after. Atypicality is the average of the two divergences: how far the filing’s language sits from its industry’s, looking both ways. Lead is their difference: positive when the filing is closer to what its class filed afterwards than to what it had filed before, so that the industry moved toward it. Within a class and year the two are nearly uncorrelated ($r = -0.10$). Both are used here as percentiles within Nice class and registration year, so an effect is the difference between the top and the bottom of that distribution.

The theme model is fitted once on all classes together. Under such a model a filing that carries a theme common in other classes but rare in its own registers as atypical, and a model fitted within each class does not register it; the survival advantage of atypicality is largely a reading of such imported themes (Silver, 2026). That is the content of atypicality in this paper: language that brings to a class what other industries describe and this one rarely does. It is close to what the blue-ocean literature calls reconstructing market boundaries by looking across industries (Kim & Mauborgne, 2004).

2.3 Emergent themes and forced vocabularies

Two kinds of content measure enter the analysis, and they answer different questions. *Emergent* measures come from the fifty themes the topic model found without guidance: atypicality and lead themselves; each filing’s primary theme, the theme its description draws on most; whether that theme was imported from another class; how volatile the theme’s share of its class has been; how much of its growth came from first-time filers; the turnover of the class’s whole theme mix; and the geographic groups of Section 9, which are the filers of one primary theme in one class. Emergent themes cover every filing and impose no prior view of which products matter, but they are broad; a theme covers many products.

Forced measures are curated lists of terms matched in the description (Appendix B): the vocabularies of general-purpose technologies (internet, mobile apps, cloud, social media, artificial intelligence, blockchain), of named inventions (3D printing, drones, virtual reality, electric vehicles, gene therapy, nanotechnology), of emergent consumer categories (e-cigarettes, energy drinks, kombucha, cold brew, plant-based milk, hard seltzer, CBD), and of kinds of business (site-based, contract or subscription, platform). Forced vocabularies name products a reader recognizes and can be dated: a vocabulary *arrives* in a class in the first year at least five of the class’s filings use it, which is what the pioneer-and-follower comparison of Section 5 requires. They are chosen with hindsight, because these products became important, so they describe the fate of entrants into markets that turned out to matter, not a random draw of new products.

2.4 Factors and estimation

Table 1 lists the twenty-four factors, their source and their frequency. Every model is a linear probability model of passing the proof with class $\times$ registration-year fixed effects, so each comparison is within an industry and year; the controls are log description length, log owner filing count and foreign-basis flags; and standard errors are clustered by owner, because an owner files many marks and abandons them together. Each factor enters three times: as a main effect, and interacted with atypicality and with lead, so that it can change what either position is worth. Factors are centred, so the atypicality and lead effects are those of a registration of average composition. Significance stars use Holm-adjusted p -values within each panel of each column.

Table 1: The factors, where each comes from, and how common it is

Factor	Source	Share	Definition
Represented by counsel	Record	78%	An attorney of record on the application.
Owner files patents	Record	20%	The owner, matched by name, holds US patents (PatentsView).
Owner’s first-ever filing	Record	33%	The owner’s first filing in the record.
Owner has 25+ earlier filings	Record	20%	The owner had filed 25 or more marks before this one.
Intent-to-use filing	Record	41%	Filed on an intent-to-use basis rather than on use in commerce.
Foreign owner (not China)	Record	18%	Owner domiciled outside the US, other than China.
Owner in China	Record	3%	Owner domiciled in China.
Site-based offering (stores, restaurants, clinics)	Forced	9%	Description names a physical site: store, restaurant, hotel, clinic, salon, gym and similar.
Contract or subscription offering	Forced	5%	Description names a subscription, membership, account, insurance, banking or leasing.
Platform or network offering	Forced	2%	Description names a platform, marketplace, social network, online community or peer-to-peer service.
General-purpose-technology vocabulary	Forced	15%	Internet, mobile-app, cloud, social-media, artificial-intelligence or blockchain vocabulary.
New-invention vocabulary	Forced	0.4%	3D printing, drone, virtual-reality, electric-vehicle, gene-therapy or nanotechnology vocabulary.
Emergent consumer-category vocabulary	Forced	0.4%	E-cigarette, energy-drink, kombucha, cold-brew, plant-milk, hard-seltzer or CBD vocabulary.
Primary theme imported from another class	Emergent	5%	The filing’s primary theme held under 1% of its class in the five prior years and at least 5% of another class.
Theme-share volatility (per SD)	Emergent	SD	Year-to-year variability of the primary theme’s share of its class, over the filing year ± 5 .
Theme growth from first-time filers (per SD)	Emergent	SD	Share of the primary theme’s filings in its class over the next five years made by first-time filers.
Class theme-mix turnover (per SD)	Emergent	SD	Change in the class’s theme mix between the five years before and after the filing year.
Theme’s geographic co-location (per SD)	Geography	SD	Co-location index of the class’s US filers of the same primary theme within two years of the filing (Section 7).
Theme has a geographic hub	Geography	28%	That group of filers has at least one hub (Section 7).
Filer located in a hub	Geography	5%	The filer is located in such a hub.
Class filing-volume growth (per SD)	Record	SD	Growth in the class’s filing volume, five years after against five years before.
Filing year (per SD, about 5 years)	Record	SD	Filing year.

Factor	Source	Share	Definition
Filed in a boom (1998-2000, 2005-07)	Record	19%	Filed in 1998–2000 or 2005–07.
Filed in a bust (2001-02, 2008-09)	Record	18%	Filed in 2001–02 or 2008–09.

Record: a field of the filing or the owner’s filing history. Forced: a curated list of terms matched in the description (Appendix B). Emergent: derived from the fifty themes the topic model found without guidance. Geography: from the applicant’s ZIP code. Share: percentage of the 3.10 million registrations with the factor; SD marks continuous factors.

3 Unusual language pays and early language costs, at every stage after registration

The two positions move survival in opposite directions (Figure 1). Survival at the proof rises from 45.4% in the most typical tenth of a class and year to 50.9% in the most unusual, and falls from 48.7% in the most lagging tenth to 45.6% in the most leading. In the regression with class-by-year fixed effects and controls (Table 3, column 1), the most unusual filing passes the proof 5.4 points more often than the most typical, and the most leading 2.2 points less often than the most lagging. The two effects do not depend on each other: the interaction is -0.4 points and not distinguishable from zero.

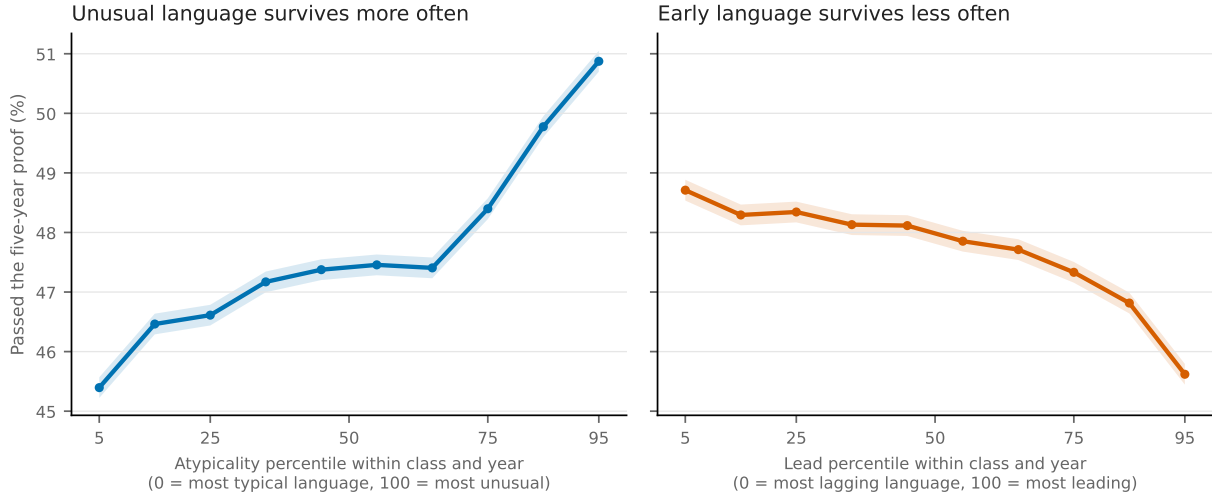


Figure 1: Share of registrations passing the five-year proof by tenth of atypicality (left) and of lead (right), both measured within Nice class and registration year. Survival is adjusted for class $\times$ registration-year fixed effects; bands are 95% intervals. 3.10 million registrations, 2002–2018.

Both effects persist at the ten-year renewal. Among registrations that passed the proof, the most unusual is renewed 4.3 points more often than the most typical and the most leading 0.9 points less often than the most lagging. The advantage of unusual language keeps most of its size; the cost of early language keeps less than half.

Unusual language is harder to register and early language easier. Among first-time filers, one standard deviation of atypicality lowers the chance of registration by 1.0 point, and one standard deviation of lead raises it by 0.5 (Table 2). Examination screens against unusual descriptions and

passes descriptions that match where the class is going; the market then reverses both.

Table 2: The two positions at each stage (first-time filers)

Stage (points per standard deviation)	Atypicality	Lead
Registered, of all filed	-0.98	+0.47
Passed the five-year proof, of registered	+1.31	-0.97
Renewed at year ten, of those that passed	+1.38	-0.43
Filed a private-offering notice, of registered	-0.17	-0.32
Listed after the offering, of those that filed one	+0.87	-0.53

One row per owner (the owner's first scored filing); class $\times$ first-filing-year fixed effects; log description length held.
All coefficients differ from zero at $p < 0.001$.

Capital markets agree with the proof. Owners whose first registration led are less likely to file a private-offering notice (-0.32 points per SD, on a base of 2.5%) and, among those that did, less likely to list (-0.53 on a base of 3.5%). Unusual first registrations are slightly less likely to raise private money and more likely to list once they have.

4 The magnitude of each factor

Counsel is the largest single factor, larger than either position. Figure 2 places every factor on one scale, each estimated alone with atypicality and lead. Registrations with an attorney of record pass the proof 18.7 points more often than those without. The next largest are the owner's filing history (owners with 25 or more earlier marks pass 12.5 points less often, because large portfolios let more marks lapse; first-time filers 5.1 less often), the filing basis (intent-to-use filings 6.0 less often), and what the filing describes: emergent consumer categories pass 13.9 points less often, new-invention vocabulary 7.8, platform businesses 8.9, and general-purpose-technology vocabulary 5.4. Site-based businesses pass 2.0 points more often. Owners who patent pass 3.0 points less often; as with large portfolios, patenting marks a larger firm with more marks to let go.

Half of the raw cost of leading is composition. Adding the factors lowers the cost of leading from 2.2 points to 1.2 (Table 3, columns 1 and 3): leading filings are more often self-filed, more often platforms and more often in general-purpose technology vocabulary, and each of these survives less. The advantage of atypicality changes less, from 5.4 to 4.5 points. Seventeen of the twenty-four factors have at least one term that survives the adjustment for multiple testing (column 2); dropping the other seven moves the atypicality and lead effects by about a tenth of a point.

Where a factor changes the value of a position, it is more often the value of atypicality than the cost of lead. Eight factors change the atypicality effect after adjustment; four change the lead effect (Table 3, panels B and C). Sections 5 and 6 take them in turn.

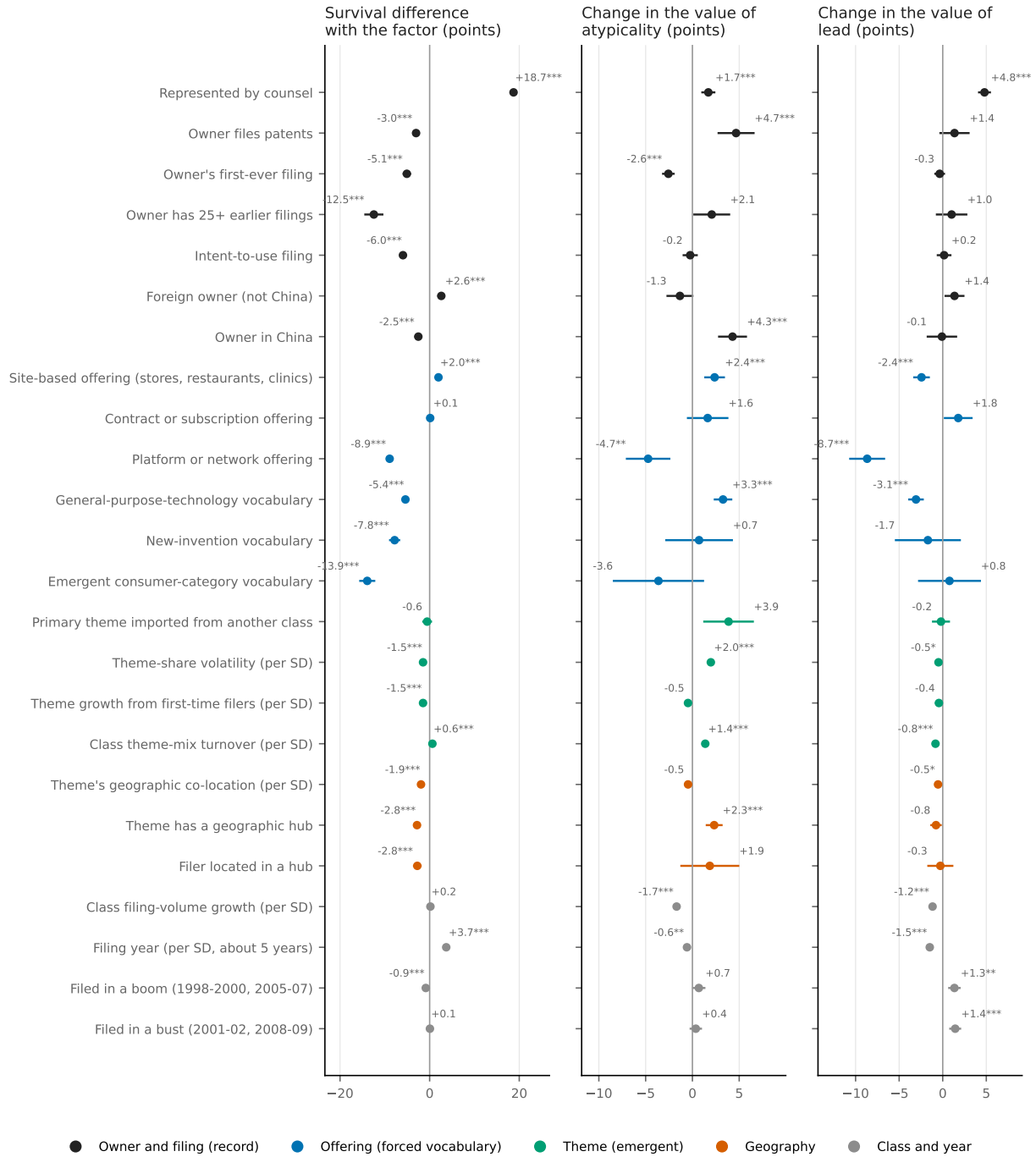


Figure 2: Each factor estimated alone, with atypicality, lead and their interaction and the controls. Left: difference in the share passing the five-year proof between registrations with and without the factor (continuous factors: per standard deviation). Centre and right: how much the factor adds to or subtracts from the atypicality effect and the lead effect (both measured as top minus bottom of the class-and-year distribution). Lines are 95% intervals; stars are Holm-adjusted within each panel (* < 0.05, ** < 0.01, *** < 0.001). Colour marks where the factor comes from (Table 1).

Table 3: Passing the five-year proof: atypicality, lead and the factors that change what each is worth

	(1) Atypicality and lead only	(2) With significant factors	(3) With all factors
A. Survival difference (percentage points)			
Atypicality (most unusual – most typical)	+5.43*** (0.24)	+4.61*** (0.21)	+4.48*** (0.22)
Lead (most leading – most lagging)	–2.21*** (0.21)	–1.15*** (0.23)	–1.18*** (0.24)
Represented by counsel		+18.70*** (0.16)	+18.71*** (0.16)
Owner files patents		–1.85*** (0.38)	–1.86*** (0.38)
Owner’s first-ever filing		–1.09 (0.60)	–1.09 (0.60)
Owner has 25+ earlier filings		–10.29*** (1.12)	–10.29*** (1.12)
Intent-to-use filing		–6.88*** (0.13)	–6.79*** (0.14)
Foreign owner (not China)			+0.99 (0.61)
Owner in China		+0.74 (0.41)	+1.62 (0.61)
Site-based offering (stores, restaurants, clinics)		+2.26*** (0.19)	+2.24*** (0.19)
Contract or subscription offering			+0.00 (0.34)
Platform or network offering		–5.59*** (0.43)	–5.60*** (0.43)
General-purpose-technology vocabulary		–4.69*** (0.16)	–4.66*** (0.16)
New-invention vocabulary		–6.39*** (0.62)	–6.42*** (0.62)
Emergent consumer-category vocabulary		–12.16*** (0.90)	–12.17*** (0.90)
Primary theme imported from another class			+0.98 (0.57)
Theme-share volatility (per SD)		–0.83*** (0.10)	–0.91*** (0.10)
Theme growth from first-time filers (per SD)		–1.14*** (0.13)	–1.13*** (0.13)
Class theme-mix turnover (per SD)		+0.68*** (0.13)	+0.60*** (0.13)
Theme’s geographic co-location (per SD)		–1.29*** (0.10)	–1.30*** (0.10)
Theme has a geographic hub		–0.46 (0.20)	–0.46 (0.24)
Filer located in a hub			+0.05 (0.52)
Class filing-volume growth (per SD)		+0.23 (0.11)	+0.27 (0.12)
Filing year (per SD, about 5 years)			+0.48 (0.32)
Filed in a boom (1998-2000, 2005-07)			–0.37 (0.19)
Filed in a bust (2001-02, 2008-09)			–0.09 (0.17)
B. Change in the value of atypicality (points)			
Lead	–0.37 (0.54)	–1.10 (0.59)	–0.75 (0.59)
Represented by counsel		+0.20 (0.35)	+0.25 (0.36)
Owner files patents		+3.79*** (0.90)	+3.91*** (0.89)
Owner’s first-ever filing		–1.74*** (0.27)	–1.68*** (0.28)
Owner has 25+ earlier filings		–0.65 (0.79)	–0.76 (0.80)
Intent-to-use filing		–0.22 (0.34)	–0.30 (0.35)
Foreign owner (not China)			–1.53 (0.78)
Owner in China		+6.01*** (0.78)	+5.88*** (0.79)
Site-based offering (stores, restaurants, clinics)		+2.67*** (0.55)	+2.74*** (0.56)
Contract or subscription offering			–0.80 (1.16)
Platform or network offering		–4.14** (1.24)	–3.76 (1.29)

	(1)	(2)	(3)
General-purpose-technology vocabulary	+1.99** (0.54)	+2.13*** (0.48)	
New-invention vocabulary	-3.51 (1.77)	-3.08 (1.78)	
Emergent consumer-category vocabulary	-1.46 (2.48)	-1.64 (2.48)	
Primary theme imported from another class		+0.85 (1.41)	
Theme-share volatility (per SD)	+0.81*** (0.20)	+0.37 (0.22)	
Theme growth from first-time filers (per SD)	+0.27 (0.22)	+0.15 (0.25)	
Class theme-mix turnover (per SD)	+0.21 (0.19)	+0.32 (0.18)	
Theme's geographic co-location (per SD)	-1.77*** (0.22)	-1.72*** (0.23)	
Theme has a geographic hub	+2.86*** (0.53)	+2.12*** (0.52)	
Filer located in a hub		+1.62 (1.65)	
Class filing-volume growth (per SD)	-1.54*** (0.19)	-1.53*** (0.19)	
Filing year (per SD, about 5 years)		-0.30 (0.21)	
Filed in a boom (1998-2000, 2005-07)		+0.03 (0.41)	
Filed in a bust (2001-02, 2008-09)		-0.63 (0.43)	
C. Change in the value of lead (points)			
Represented by counsel	+5.17*** (0.36)	+5.05*** (0.37)	
Owner files patents	+0.33 (0.73)	+0.29 (0.73)	
Owner's first-ever filing	+0.73 (0.26)	+0.72 (0.27)	
Owner has 25+ earlier filings	-0.15 (0.69)	-0.07 (0.69)	
Intent-to-use filing	-0.82 (0.35)	-0.97 (0.36)	
Foreign owner (not China)		+0.62 (0.63)	
Owner in China	+0.68 (0.84)	+1.37 (0.86)	
Site-based offering (stores, restaurants, clinics)	-1.26 (0.47)	-1.33 (0.47)	
Contract or subscription offering		+1.92 (0.82)	
Platform or network offering	-4.87*** (1.08)	-4.88*** (1.09)	
General-purpose-technology vocabulary	-2.11*** (0.45)	-2.12*** (0.44)	
New-invention vocabulary	-1.27 (1.90)	-1.00 (1.92)	
Emergent consumer-category vocabulary	-0.93 (1.85)	-0.55 (1.85)	
Primary theme imported from another class		-0.56 (0.58)	
Theme-share volatility (per SD)	+0.42 (0.20)	+0.37 (0.21)	
Theme growth from first-time filers (per SD)	+0.01 (0.23)	+0.14 (0.23)	
Class theme-mix turnover (per SD)	-0.65*** (0.16)	-0.77*** (0.18)	
Theme's geographic co-location (per SD)	-0.24 (0.18)	-0.27 (0.19)	
Theme has a geographic hub	-0.28 (0.38)	+0.06 (0.46)	
Filer located in a hub		+0.03 (0.77)	
Class filing-volume growth (per SD)	-0.60* (0.19)	-0.34 (0.21)	
Filing year (per SD, about 5 years)		-0.55 (0.25)	
Filed in a boom (1998-2000, 2005-07)		+0.64 (0.47)	
Filed in a bust (2001-02, 2008-09)		+0.16 (0.42)	

	(1)	(2)	(3)
Registrations	3,104,534	3,104,534	3,104,534
Factors	0	17	24

Linear probability models of passing the five-year proof of continued use, registrations 2002–2018. Atypicality and lead are percentiles within Nice class and registration year, centred, so a coefficient in panel A is the difference between the top and the bottom of the class-and-year distribution. Factors are centred (binary factors 0/1, continuous factors in SD units), so the atypicality and lead rows are the effects for a registration of average composition. Panel B reports how much a factor adds to or subtracts from the atypicality effect; panel C, from the lead effect. Column 2 keeps the factors with at least one term significant in column 3. All models: class $\times$ registration-year fixed effects; log description length, log owner filing count and foreign-basis flags; standard errors clustered by owner in parentheses. Stars: Holm-adjusted p within each panel of each column; * < 0.05, ** < 0.01, *** < 0.001.

5 The value of unusual offerings, and the imitation that erodes it

A blue-ocean effect exists, and it lasts. With every factor held, the most unusual filing in its class and year passes the proof 4.5 points more often than the most typical, and among those that pass it is renewed 4.3 points more often. Because atypicality under a single theme model registers themes imported from other classes (Section 2), the effect belongs to filings that describe, in their class, what other industries describe. Flagging imported primary themes directly adds nothing once atypicality is held (+1.0 points, not significant): the advantage runs through how unusual the whole description is, not through a single borrowed theme.

Growth in the class’s filing volume erodes it. Each standard deviation of five-year growth in the class’s filings reduces the atypicality advantage by 1.5 points: it is worth 6.0 points one SD below the mean and 3.0 one SD above. Figure 3b shows where the loss falls. In the fastest-growing third of class-years, survival is nearly flat from the 10th to the 80th percentile of atypicality, about one point above the most typical tenth, and rises only in the most unusual fifth. In the slowest-growing third it rises across the whole range.

Geographic concentration erodes it more. Each standard deviation of a theme’s co-location reduces the atypicality advantage by 1.7 points: from 6.2 points one SD below the mean to 2.8 one SD above. In the raw comparison (Figure 3c), unusual filings in the most dispersed third of themes pass the proof 6.9 points more often than typical ones; in the most concentrated third, 3.9. Section 9 returns to this.

A filing’s own lead does not change the advantage detectably. The most lagging third of filings gains 6.4 points from the most typical to the most unusual tenth and the most leading third 4.7 (Figure 3a), but the interaction in the regression is -0.8 points and not distinguishable from zero, at the proof or at renewal (-1.4 , s.e. 0.9). What erodes the advantage is growth in the class and concentration of the theme, not the filing’s own position ahead of its class.

First-time filers keep less of it. For an owner’s first filing the atypicality advantage is 3.4 points; for later filings, 5.0.

Pioneers of a named product fail, and so do its early followers. The forced vocabularies can be dated. Filings using a vocabulary in its arrival year or the year after pass the proof 12.8 points less often than the other filings of their class and year; filings two to five years after arrival, 11.8 points less often; later filings, 6.0 points less often. The first two do not differ ($t = 0.5$). Pioneers fail, as Golder & Tellis (1993) found in fifty product histories; the early followers here fail at the same rate.

Leading costs most where markets grow and technology turns over together. Dividing class-years



Figure 3: Share passing the five-year proof by tenth of atypicality, for (a) the most lagging and most leading third of filings in their class and year; (b) class-years in the slowest- and fastest-growing third of filing volume (five years after against five years before); (c) themes in the most dispersed and most concentrated third of geographic co-location (Section 9). Survival adjusted for class $\times$ registration-year fixed effects; bands are 95% intervals.

at the median of theme-mix turnover and of filing-volume growth, and holding a filing-year trend, leading costs 4.7 points where both change fast, 2.7 where only volume grows, 1.0 in calm class-years and 0.8 where only the theme mix turns over. The pace-of-change account predicts the most durable first-mover advantage where the market grows and technology is stable (Suárez & Lanzolla, 2007); the record puts a larger cost there than in calm class-years.

6 What lets the creator keep the value: protection and complementary assets

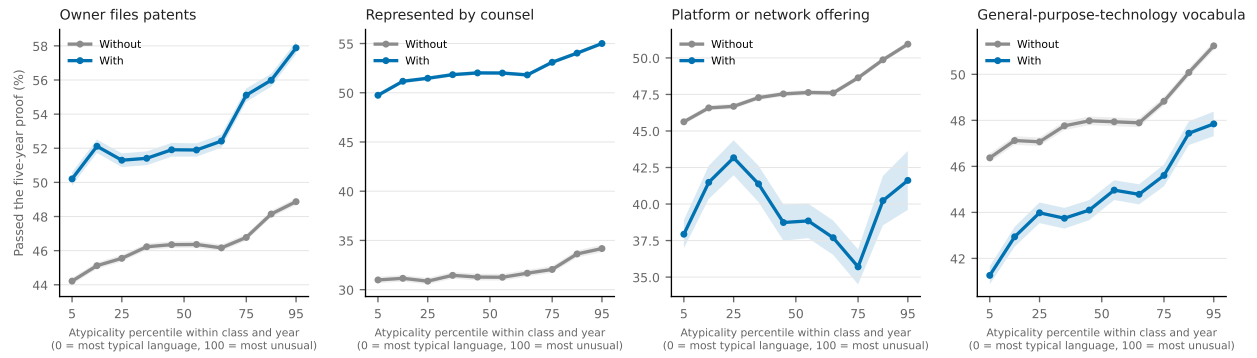
Patents double the advantage of unusual language. For owners who patent, the most unusual filing passes the proof 7.6 points more often than the most typical; for owners who do not, 3.7 (Table 3, panel B, +3.9 points; Figure 4a). Patents do not change the cost of leading (+0.3, not significant). Owners who can protect the technology behind an unusual offering keep more of what it earns; the protection does not extend to being early.

Site-based businesses, general-purpose technologies and owners in China also keep more of it. The atypicality advantage is 7.0 points for site-based businesses (4.2 for others), 6.3 for filings in general-purpose-technology vocabulary (4.2), 6.0 for themes with a geographic hub (3.9), and 10.2 for owners in China (4.3). For owners in China the typical filing fares badly and the unusual one does not.

Platform businesses do not keep it. For platform and network offerings the atypicality advantage is 0.8 points, against 4.6 for other filings; the difference (-3.8) does not survive the adjustment for multiple testing, but it is the largest negative change in panel B.

Counsel confines the cost of leading to the most leading filings. Among counsel-represented registrations, survival is flat across the first nine tenths of lead (52.1–52.6%) and 1.7 points lower in the most leading tenth; among self-filed registrations it falls steadily, from 34.0% in the most lagging tenth to 29.3% in the most leading (Figure 4b). In the linear specification of Table 3 the

a. The value of unusual language



b. The cost of early language

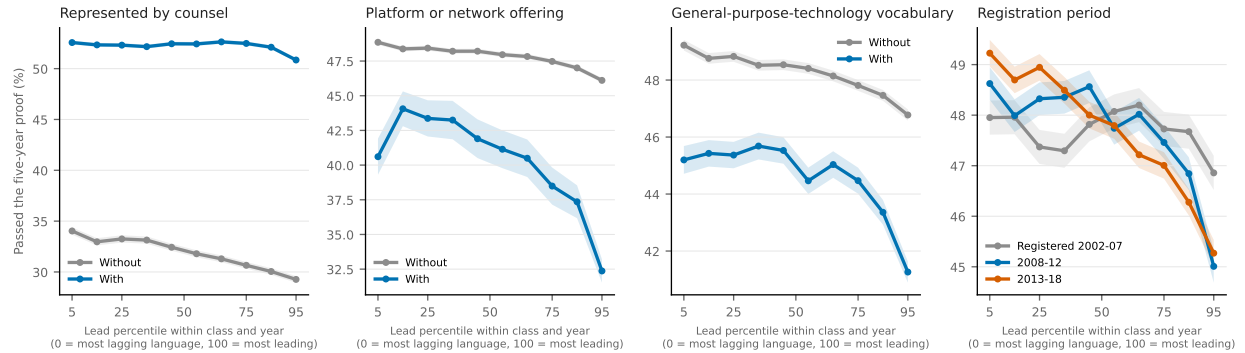


Figure 4: Share passing the five-year proof by tenth of atypicality (a) and of lead (b), for registrations with and without each factor, and (bottom right) by registration period. Survival adjusted for class $\times$ registration-year fixed effects; bands are 95% intervals.

counsel-represented slope is -0.1 points, because a drop confined to one tenth barely moves a slope; the quintile contrast with the same controls is 1.15 points with counsel and 1.80 without (Silver, 2026). Counsel does not change the value of unusual language ($+0.3$). At renewal, counsel still halves the cost of leading ($+2.2$ points).

Platforms and general-purpose technologies pay more for leading. Leading costs platform businesses 6.0 points (1.1 for other filings) and filings in general-purpose-technology vocabulary 3.0 (0.9). The platform penalty persists at renewal (-8.5 against other filings); the extra cost for general-purpose technology is paid at the proof and not at renewal (-0.6 , not significant).

The cost of leading has grown over time. For registrations of 2002–07 the most leading filing passed the proof as often as the most lagging (-0.1 points); for 2008–12 and 2013–18, 2.2 and 2.0 points less often (Figure 4b, right). Once a filing-year trend is held, booms (1998–2000, 2005–07) and busts (2001–02, 2008–09) do not differ from other years.

7 Who collects within a wave of entry

Pioneers and early followers fail alike. Within each class, a curated vocabulary arrives in the first year at least five of the class’s filings use it. Registrations using a new vocabulary in its first two years in the class are out of use at the five-year declaration 12.8 points more often than other registrations of the same class and year; those two to five years in, 11.8 points more often; those six or more years in, 6.0. The first two groups do not differ ($t = 0.5$). The pioneers of these markets fail, as Golder & Tellis (1993) found in named histories, but so do the firms that follow them in the next few years; the cost belongs to joining the wave early, not to being first in it.

Buyers do not favour early entrants. A business can also be rewarded by being bought. Recorded transfers of trademark ownership between companies show that 0.31% of first-time owners had their first mark bought by another company, almost twice the share that listed. The most leading fifth of first registrations were bought slightly less often than the most lagging (0.29% against 0.33%, $t = -2.1$), and bought by an established firm, one holding 25 or more marks, two-thirds as often (0.046% against 0.069%, $t = -3.1$).

8 Industry structure and the cost of being early

9 Geographic clustering

A measure of clustering needs to allow several hubs and to net out where firms are anyway. Distance to the nearest neighbour or to a single centre cannot recognize an industry split between the Bay Area and New York, and every industry’s filers concentrate where people and firms are. We place each US applicant at the Census centroid of the ZIP code on its application (re-parsed from the annual case-file archives; 99.8% of US owners placed, 92% at their own ZIP code) and measure a group of N filers by the average, over all pairs, of a distance kernel:

$$C = \frac{1}{N(N-1)} \sum_{i \neq j} k(d_{ij}), \quad k(d) = \max\left(0, 1 - \frac{d}{200 \text{ miles}}\right).$$

Two filers in the same place score 1, two 100 miles apart 0.5, two more than 200 miles apart 0; C is the chance that two filers drawn at random are neighbours, weighted by how near. Any number of hubs contributes: a group split evenly between two distant cities scores half of what it would in one. The same index for all filers in the same classes and years gives the baseline B , and co-location

is $L = (C - B)/(1 - B)$: zero for a group placed like its classes, one when every filer is in one place. This is the distance-based comparison with a counterfactual of Duranton & Overman (2005), summarized at one bandwidth. A *hub* is a place near which (kernel-weighted) at least 10% of the group's filers sit, at least 1.2 times the share of all filers there; hubs are taken in order of their share, each excluding places within 200 miles of the last.

The measure recognizes industries with known geography (Figure 5, Table 4). Semiconductor filers of 1995–2005 have co-location 0.20, with 48% of them near Santa Clara against 12% of all filers in their class. Social-networking filers of 2004–12 have three hubs, New York, San Francisco and Los Angeles. Biotechnology filers cluster near Philadelphia, the Bay Area and New Haven; casino filers near Las Vegas; country-music filers near Nashville. Broad vocabularies used across the economy, such as internet language in 1998–2001, show almost no co-location ($L = 0.01$).

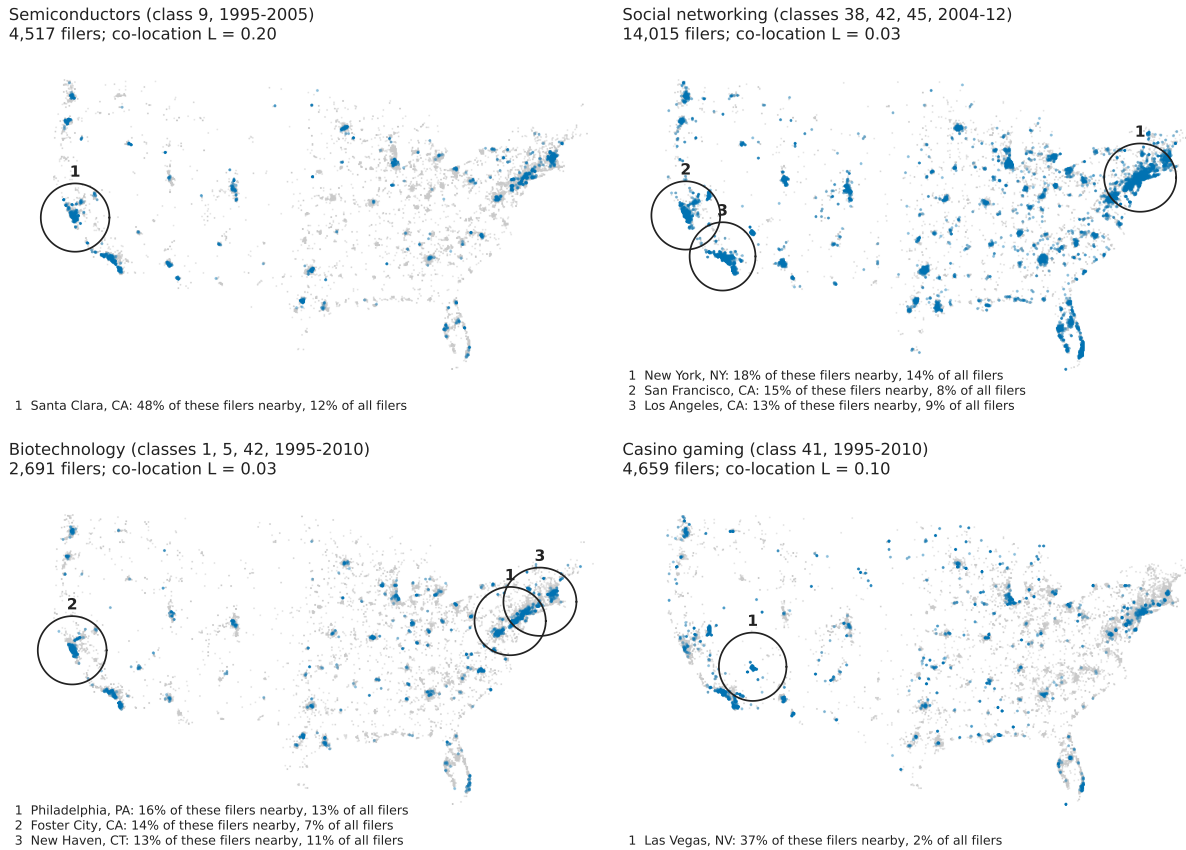


Figure 5: US filers whose descriptions use each vocabulary (blue), against a sample of all filers in the same classes and years (grey). Circles are the detected hubs, 200 miles in radius; the key gives the share of the group and of all filers near each.

Clustering lowers survival and erodes the advantage of unusual language; it does not change the cost of leading. For the regressions, each registration is assigned the co-location of its group: the class's US filers of the same primary theme within two years of its filing. Each standard deviation of co-location lowers survival by 1.3 points and the atypicality advantage by 1.7 points; it changes the lead effect by -0.3 , not distinguishable from zero. A group with an identifiable hub keeps more of the atypicality advantage ($+2.1$ points), and a filer located in a hub fares no differently from one

Table 4: The co-location measure on groups with known geography

Filers using the vocabulary of	Filers	L	Hubs: share of the group nearby / share of all filers
internet vocabulary, classes 9/35/38/42, 1998-2001	60,903	0.014	San Francisco, CA 13% / 10%
semiconductors, class 9, 1995-2005	4,517	0.203	Santa Clara, CA 48% / 12%
wine, class 33, 2000-2010	18,799	0.056	Napa, CA 35% / 26%
country music, class 41, 1995-2010	163	0.041	Nashville, TN 28% / 1%
casino gaming, class 41, 1995-2010	4,662	0.095	Las Vegas, NV 37% / 2%
biotechnology, classes 1/5/42, 1995-2010	2,695	0.031	Philadelphia, PA 16% / 13%; Foster City, CA 14% / 7%; New Haven, CT 13% / 11%
social networking, classes 38/42/45, 2004-2012	14,041	0.033	New York, NY 18% / 14%; San Francisco, CA 15% / 8%; Los Angeles, CA 13% / 9%
online advertising, classes 35/42, 2005-2015	12,028	0.027	New York, NY 17% / 14%; San Francisco, CA 14% / 7%; Los Angeles, CA 12% / 9%
software as a service, class 42, 2008-2016	20,783	0.008	San Francisco, CA 15% / 10%
hip hop, classes 9/25/41, 1995-2010	507	0.058	New York, NY 29% / 16%

L : co-location of the group net of all filers in the same classes and years (0 = placed like the classes, 1 = all in one place). Nearby: kernel-weighted within 200 miles.

outside it, in survival (+0.1) or in either effect. Concentration in place reduces what an unusual description is worth; being in the hub itself neither helps nor hurts.

10 Industries: where the cost of leading falls and whether it lasts

Industries differ in a stable way. Estimated class by class, the lead effect at the proof and the lead effect at renewal correlate at 0.67 across classes, and the atypicality effects at 0.74 (Figure 6). In twelve classes leading costs at least 2 points at both tests: most in tobacco and vaping (11.3 points at the proof, 7.9 at renewal) and technology and science services (9.6 and 7.9), then medical devices, meat and dairy, lighting and heating, hand tools and fabrics. In five it costs only at the proof: electronics and software, advertising and business services, hotels and restaurants, chemicals and clothing. In four it costs only at renewal: education and entertainment, housewares, jewelry and fuels. Leading is rewarded at both tests in construction and repair (4.0 and 4.5), transport and storage, toys and sporting goods, firearms and treatment of materials, and at the proof only in beer and soft drinks (9.6), paints and legal services. Appendix A lists every class.

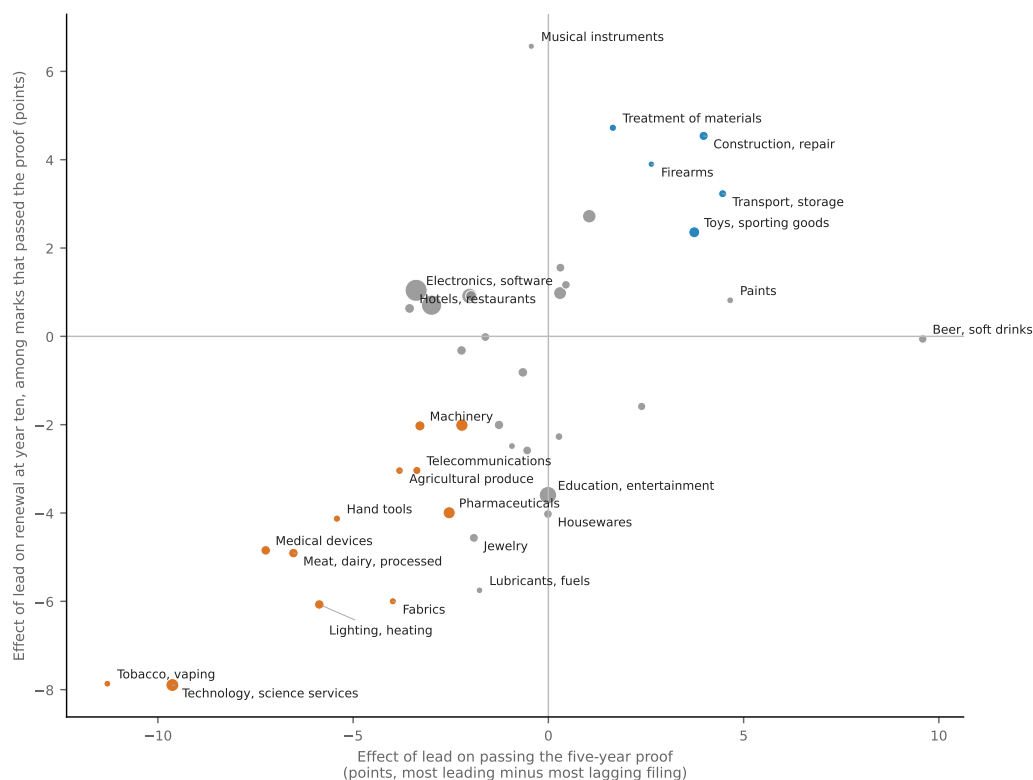


Figure 6: The lead effect by Nice class at the five-year proof and at renewal among registrations that passed it (top minus bottom of the class-and-year distribution; class $\times$ registration-year fixed effects and controls). Orange: leading costs at least 2 points at both tests; blue: leading pays at least 1.5 points at both. Point size is proportional to registrations.

Tobacco's penalty is a change of product. The class-34 penalty is confined to registrations of 2013–18 (−23.4 points; +4.1 and +0.5 in 2002–07 and 2008–12), when e-cigarette and vaping language went from none of the class's registrations to 46%. Vaping registrations pass the proof 15.8 points less often than the class's others, and within each product — vaping, cigars, cigarettes,

hookah — leading is not penalized. In class 34 leading language meant vaping, and vaping marks were abandoned.

Beer's reward belongs to the craft-beer decade. In class 32 the rising product was beer, which went from a third of the class's registrations in 2002–07 to two-thirds in 2013–18. Beer registrations pass the proof 7.5 points more often than the class's others, and leading beer filings 10.1 points more often than lagging ones; at renewal leading neither helps nor hurts. Across the classes where leading pays, survival is highest near the 65th percentile of lead and lower at both ends: the cost there falls on lagging filings, and the most leading tenth does not gain.

Where leading pays, unusual language does not. In the classes where leading costs least, the most unusual filings pass the proof 2.9 points less often than the most typical; where it costs most, atypicality has no effect (+0.7); in the large middle it is worth 5.8 points. Unusual language pays most in electronics and software (+16 points), advertising and business services (+13), telecommunications (+10) and pharmaceuticals (+11), and costs in machinery (−11), legal and personal services (−7), chemicals and paints (−5).

11 Discussion: who profits

A new kind of offering creates value in this record, and the value goes to its creator only under the conditions the appropriability account names. Offerings that bring to an industry what other industries describe are still in use more often, and the advantage lasts to renewal; it is kept where it can be protected by patents and lost where entry grows and rivals concentrate. Arriving where an industry is about to go is a different matter: early entrants lose value on average, the loss falls on those without the complementary assets that counsel and an established portfolio stand for, and neither patents nor network businesses protect against it.

Comparing the forces on one scale

The literatures on entry order are usually read as a contest between claims. Measured together on one outcome, the forces they name are all present, and their sizes differ by an order of magnitude. Retaining counsel is worth 18.7 points of survival at the five-year proof. Being unusual for one's industry is worth 4.5 at the proof and 4.3 at renewal. Being early costs 1.2 for the average registration, and 5 to 11 points for self-filed registrations, platform businesses and a dozen industries.

The blue-ocean effect is the larger of the two positions, and it is contingent on competition. Language that brings to a class what other industries describe survives more often, as the practitioner literature holds (Kim & Mauborgne, 2004). The advantage is halved where the class's filing volume grows by a standard deviation and where the filers of the same theme concentrate in place, and in fast-growing class-years only the most unusual filings keep it. Uncontested space pays while it stays uncontested; growth in entry and concentration of rivals are what the record shows removing the advantage.

Protection is specific. Patents double the advantage of unusual language and do nothing for the cost of being early, consistent with the appropriability account, in which protection covers a technology rather than a position in time (Teece, 1986). Counsel confines the cost of being early to the most leading filings and does nothing for the value of being unusual. Platform businesses keep neither: they pay the most for leading and keep the least of the atypicality advantage, the reverse of the network-effects argument for first movers (Lieberman & Montgomery, 1988).

Clusters cut both ways. Concentration of a theme's filers erodes the value of unusual language, while a theme with an identifiable hub keeps more of it. The record cannot separate imitation

from shared suppliers, labour or demand; it can say that the two measures move the advantage in opposite directions and by comparable amounts.

What the measures cannot say. Lead and atypicality measure the position of a filing’s language relative to its industry’s, not the novelty of a product, and a mark can lead because others imitated it, because all responded to the same change, or by coincidence. The five-year proof records whether goods were still sold under the mark, not profits. The forced vocabularies were chosen with hindsight. The co-location measure uses the applicant’s address, which for a large firm may be a headquarters rather than where the product is made. The estimates are associations within industry and year, with the controls stated; none identifies a causal effect of choosing unusual or early language.

A Every class

Lead and atypicality effects by Nice class (percentage points, top minus bottom of the class-and-year distribution; standard errors for lead in parentheses), at the five-year proof (registrations 2002–18) and at renewal among those that passed (2002–13). A class is placed in a lead pattern when its effect at a test is at least 2 points (1.5 for persistent rewards).

Class		Registrations	Lead		Atypicality	
			Proof	Renewal	Proof	Renewal
<i>Lead: penalty persists</i>						
34	Tobacco, vaping	13,993	−11.3 (2.8)	−7.9 (4.5)	+5.1	−3.4
42	Technology, science services	127,407	−9.6 (0.6)	−7.9 (1.1)	+5.7	+3.3
10	Medical devices	55,537	−7.2 (1.0)	−4.8 (1.5)	−1.6	−2.1
29	Meat, dairy, processed	55,503	−6.5 (1.2)	−4.9 (1.8)	−0.5	−1.7
11	Lighting, heating	55,383	−5.9 (1.1)	−6.1 (1.7)	−1.0	+5.2
8	Hand tools	19,542	−5.4 (1.6)	−4.1 (2.6)	−2.4	−4.2
24	Fabrics	18,531	−4.0 (1.8)	−6.0 (2.9)	+4.2	+4.5
31	Agricultural produce	25,317	−3.8 (1.8)	−3.0 (2.7)	+6.8	+0.8
38	Telecommunications	32,421	−3.4 (1.3)	−3.0 (2.2)	+10.0	+11.6
7	Machinery	66,181	−3.3 (1.0)	−2.0 (1.3)	−11.2	−8.8
5	Pharmaceuticals	108,181	−2.5 (1.4)	−4.0 (1.2)	+10.7	+7.6
3	Cosmetics, cleaning	113,917	−2.2 (0.9)	−2.0 (1.4)	+8.1	+11.1
<i>Lead: penalty appears late</i>						
14	Jewelry	45,289	−1.9 (1.0)	−4.6 (2.0)	+0.7	−4.1
4	Lubricants, fuels	11,780	−1.8 (2.6)	−5.8 (3.8)	+12.2	+1.6
21	Housewares	40,927	0.0 (1.2)	−4.0 (1.9)	+3.8	+5.1
41	Education, entertainment	254,579	0.0 (0.5)	−3.6 (0.8)	0.0	−0.9
<i>Lead: penalty fades</i>						
43	Hotels, restaurants	60,493	−3.6 (0.9)	+0.6 (1.6)	−2.0	−6.0
9	Electronics, software	426,177	−3.4 (0.5)	+1.0 (0.7)	+15.8	+14.0
35	Advertising, business	353,547	−3.0 (0.4)	+0.7 (0.7)	+12.6	+8.9
1	Chemicals	55,166	−2.2 (1.0)	−0.3 (1.4)	−5.3	−4.2
25	Clothing	197,259	−2.0 (0.5)	+0.9 (0.9)	+3.7	+4.3
<i>Lead: benefit persists</i>						
40	Treatment of materials	19,797	+1.7 (1.5)	+4.7 (2.5)	−2.3	−2.3

Class	Registrations	Lead		Atypicality	
		Proof	Renewal	Proof	Renewal
13 Firearms	8,387	+2.6 (2.6)	+3.9 (4.2)	+2.7	−4.6
28 Toys, sporting goods	82,356	+3.7 (1.9)	+2.4 (1.9)	+4.2	−1.7
37 Construction, repair	49,178	+4.0 (1.0)	+4.5 (1.6)	−0.9	−3.9
39 Transport, storage	30,844	+4.5 (1.3)	+3.2 (2.1)	+0.4	+1.3
<i>Lead: benefit fades</i>					
45 Legal, personal services	34,032	+2.4 (1.1)	−1.6 (2.4)	−7.2	−5.0
2 Paints	13,215	+4.7 (2.0)	+0.8 (3.2)	−4.8	−3.9
32 Beer, soft drinks	37,724	+9.6 (1.3)	−0.1 (2.7)	−2.2	−5.8
<i>Lead: small</i>					
30 Coffee, bakery, staples	79,726	−2.0 (0.9)	+0.9 (1.5)	−2.0	+1.8
20 Furniture	43,033	−1.6 (1.1)	0.0 (1.9)	+2.5	+5.6
33 Wine, spirits	52,838	−1.3 (1.2)	−2.0 (2.0)	−1.7	−2.0
17 Rubber, plastics	14,684	−0.9 (1.7)	−2.5 (2.5)	−4.9	−0.9
44 Medical, beauty services	56,969	−0.7 (0.9)	−0.8 (1.7)	−3.1	+0.3
6 Metal goods	41,413	−0.5 (1.0)	−2.6 (1.5)	−2.9	−1.0
15 Musical instruments	6,486	−0.4 (2.6)	+6.6 (3.9)	−9.1	+2.3
19 Building materials	25,319	+0.3 (1.5)	−2.3 (2.1)	+2.6	+5.9
16 Paper, printed matter	126,216	+0.3 (0.7)	+1.0 (1.0)	+0.3	−2.6
12 Vehicles	43,416	+0.3 (1.2)	+1.6 (1.7)	−0.3	−1.0
18 Leather goods	39,283	+0.4 (1.1)	+1.2 (2.1)	+5.9	+5.8
36 Insurance, finance	141,201	+1.0 (0.7)	+2.7 (1.1)	−0.9	+0.1

B Forced vocabularies

Each vocabulary is a list of terms matched, case-insensitively, anywhere in the goods-and-services description.

General-purpose technologies. Internet: internet, online, website, web page, world wide web, e-commerce, electronic commerce. Mobile apps: mobile app(lication), smartphone app, downloadable mobile. Cloud: cloud computing, cloud-based, software as a service, SaaS, platform as a service. Social media: social media, social network. Artificial intelligence: artificial intelligence, machine learning, deep learning, neural network, natural language processing, computer vision, chatbot. Blockchain: blockchain, cryptocurrency, bitcoin, non-fungible token, distributed ledger, virtual currency.

Inventions. 3D printing and additive manufacturing; drones and unmanned aerial vehicles; virtual and augmented reality; electric vehicles and EV charging; gene therapy, gene editing, CRISPR; nanotechnology, nanoparticles, nanomaterials.

Emergent consumer categories. Electronic cigarettes and vaping; energy drinks; kombucha; cold brew; plant-based, almond, oat and non-dairy milk; hard and alcoholic seltzer; cannabidiol, CBD, hemp-derived.

Kinds of business. Site-based: restaurant, hotel, motel, retail store, store services, clinic, salon, spa, fitness center or club, gym, bar services, cafe, car wash, day or child care, dental, veterinary, coffee shop, bakery services. Contract or subscription: subscription, membership, account(s), insurance, banking, telecommunication, leasing, financing. Platform: platform(s), marketplace, social network, online community, peer-to-peer, connecting buyers, users or people.

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